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Simultaneous feedforward recruitment of the vasti in untrained postural tasks can be restored by physical therapy

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Abstract

Background and purpose: Physical therapy rehabilitation strategies are commonly directed at the alteration of muscle recruitment in functional movements. The aim of this study was to investigate whether feedforward strategies of the vasti in people with patellofemoral pain syndrome can be changed by a physical therapy treatment program in a randomised, double blind, placebo controlled trial.

Subjects: Forty (25 female, 15 male) subjects aged 40 yrs or less (27.2 ± 7.8 yrs).

Methods: Subjects were allocated to either a placebo treatment or a physical therapy intervention program. The postural challenge used as the outcome measure was not included in the training program. Electromyography (EMG) onsets of vastus medialis obliquus (VMO), vastus lateralis (VL), tibialis anterior and soleus were assessed before and after the six week standardised treatment programs.

Results: At baseline the EMG onset of VL occurred prior to that of VMO in both subject groups. Following physical therapy intervention there was a significant change in the time of onset of EMG of VMO compared to VL with the onsets occurring simultaneously. This change was associated with a reduction in symptoms. In contrast, following placebo intervention the EMG onset of VL still occurred prior to that of VMO.

Conclusion and discussion: The results indicate that the feedforward strategy used by the central nervous system to control the patella can be restored. Importantly, the data suggest that this intervention produced a change that was transferred to a task that was not specifically included in the training program. Furthermore, the change in motor control was associated with clinical improvement in symptoms.

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Keywords: Motor control; Quadriceps; Training; Electromyography; Postural perturbation

Introduction

Changes in the pattern of activation of muscles of the knee [4,6], spine [15,16], ankle [17] and shoulder [33] have been shown to be altered in association with musculoskeletal pain. These changes have been frequently identified as delayed or reduced activity of specific muscles during postural responses that precede functional body movements [6,15,16]. As these anticipatory or feedforward postural responses occur in advance of a movement they must be preplanned by the central nervous system (CNS) (see [21] for review). Thus, changes represent a deficit in the preplanned strategy used by the CNS to prepare the body for the perturbation that results from the movement. In view of these findings of changes in control strategy, rehabilitation commonly aims to restore the normal coordination of the muscle activity.

One strategy to retrain normal coordination involves voluntary activation of the specific muscles that are affected by pain, during functional movements [23,29]. It has been shown that these interventions are associated

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with a positive clinical outcome including a reduction in pain, improved function and decreased recurrence of pain [12,26]. Although there is preliminary data that suggest that specific motor training is associated with an improvement in motor control during a task that has been trained [2], it is not known whether the improvement in coordination transfers to other, untrained tasks nor whether feedforward strategies can be trained.

In people with patellofemoral pain syndrome (PFPS) the recruitment of the vasti have been found to be altered. Pain-free subjects have simultaneous onset of activity of medial (vastus medialis obliquus, VMO) and lateral (vastus lateralis, VL) vasti during stair stepping and ankle movements in standing. In contrast, the onset of VMO occurs after that of VL in people with PFPS [4,6]. This change is consistent with a change in motor control rather than a change in muscle performance (e.g. strength or endurance, although these cannot be excluded). Physical therapy rehabilitation strategies for PFPS commonly aim to improve the timing of activation of VMO in subjects with PFPS. While such strategies have been shown to alter the motor control of the vasti in a specific task that is trained [2] it is not known whether it is possible to train feedforward strategies. Thus the aim of this study was to investigate whether a physical therapy program could change the feedforward strategy of the vasti in subjects with PFPS more so than a placebo treatment program. The clinical outcome data have been reported in full elsewhere [7].

Methods

Subjects

Forty (25 f, 15 m) of the 71 subjects taking part in a larger randomised controlled trial participated in the experiment. All were diagnosed with PFPS on the basis of clinical examination by an experienced musculoskeletal physiotherapist. Participants were included in the PFPS group if they had anterior or retropatellar knee pain reported on at least two common aggravating activities (e.g. prolonged sitting, stairs, squatting, running, kneeling and hopping/jumping). In addition they were to have pain on patellar palpation and while stepping down from a 25 cm step or during a double leg squat, have had symptoms for at least one month, have an average pain level of 3 cm or more on a 10 cm visual analogue scale (VAS), and have an insidious onset of symptoms unrelated to a traumatic incident. All participants were aged 40 years or less to reduce the likelihood of osteoarthritic changes in the patellofemoral joint. Participants were excluded if they had signs or symptoms of other pathology or exhibited features that may have affected the implementation of the trial (e.g. history of patellar taping, inability to attend a physical therapy clinic for the six week treatment program, allergic reaction to adhesive tape and lack of understanding of written and spoken English).

Subjects were randomised into a physical therapy group ($n = 22$, 13 f, 9 m) or placebo group ($n = 18$, 12 f, 6 m) using block randomisation stratified to treatment centre. The mean (SD) age, height and weight of the physical therapy group were 28.8 (8.0) yr, 171.5 (9.9) cm and 68.8 (14.3) kg, respectively. The mean age, height and weight of the placebo group were 26.3 (7.5) yr, 170.5 (6.9) cm and 74.6 (9.5) kg respectively. The mean (range) duration of symptoms in the physical therapy group was 9.77 (1.00–72.00) months and in the placebo group 11.02 (1.50–90.01) months. There were no differences between groups for these variables assessed using independent t-tests. Both subjects and the examiner (SC) were blind to intervention allocation. The study was approved by the University of Melbourne Human Research Ethics Committee. All subjects provided written informed consent.

Electromyographic recordings

At baseline and immediately following completion of the trial, electromyography (EMG) recordings of VMO, VL, tibialis anterior (TA) and soleus (Sol) were made during a task that challenges the stability of the knee. EMG activity was recorded using pairs of surface Ag/AgCl electrodes (Graphics Control Corporation, c/o Medical Equipment Services PTY.LTD, Richmond, Australia) placed over the muscle bellies of VMO, VL, TA and Sol with an inter-electrode distance of 22 mm [5]. EMG data were preamplified (10×), bandpass filtered between 20 and 500 Hz, sampled at 1000 Hz and 12-bit A-D converted (Associative Measurement Pty, Ltd, North Ryde, N.S.W., Australia).

Procedure

Subjects stood with their arms by their side, their feet approximately 20 cm apart and with equal weight on each foot. They were instructed to rise onto their toes by contracting their triceps surae muscle ('rise' task) or to rock on to their heels by lifting their toes and contracting their TA muscle ('rock' task) as quickly as possible in response to a light. Each specific task was indicated by a different colored light in a choice reaction-time task. Subjects were instructed not to remain balanced on their toes or heels at the completion of the task, nor to attempt to return to the exact initial position. Subjects were instructed to remain relaxed between trials. Data were collected for ten repetitions each of the rock and rise tasks. This procedure has previously been found to be reliable in our laboratory [5]. No subject reported pain during the tasks. Measurements of EMG onsets and pain were made by a blinded examiner (SC) before and after an intervention protocol (see below).

Pain and disability measures

Patients rated their worst pain in the last week on a 10 cm VAS [10] and completed a self-administered questionnaire of disability that is specific to the anterior knee pain (AKP) [19].

Interventions

Standardised treatment protocols were based on interventions aimed at restoration of the co-ordination of the vasti [22]. Subjects were treated once a week for six weeks by physiotherapists experienced in the treatment of PFPS.

Physical therapy intervention protocol

The physical therapy intervention program included progressive functional retraining of VMO with auditory and visual feedback of VMO EMG activity using dual channel, surface EMG biofeedback (Pathway MR-20, The Prometheus Group, Dover, USA). The specific components of the VMO retraining program were; (i) Weeks 1–2: gentle isometric contractions focussed on VMO in sitting with the knee at 90° flexion and inner range squats to 40° knee flexion combined with isometric gluteal contraction. (ii) Weeks 3–6: when the subject could complete five step downs from a 20 cm step with out pain, the step down exercise and VMO retraining with a isometric hip abduction in standing were included. In addition to the specific focussed activation of VMO, subjects were given three other interventions aimed at improving vasti function. Namely, the application of tape to the patella to produce a medial glide to the patella which has been shown to restore vasti recruitment to normal in PFPS subjects [3], stretching of the lateral patellar retinacular structures, hamstrings and anterior hip structures (which may compromise patellar tracking (and thus VMO function)) and strengthening of the hip external rotators (Gluteus medius) to improve the biomechanical alignment of the patellofemoral joint for VMO function [22].

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Placebo intervention protocols

The placebo intervention program was designed to control for the effect of interaction with a caring practitioner, but have minimal effect on the knee. It consisted of placebo application of tape to the knee (vertical taping using non-rigid tape applied with the knee at 90° such that the alignment of the patella was not altered and its movement not affected when the subject completed the perturbation task), sham therapeutic ultrasound (custom made devices, Metron Australia) and the light application of a non-therapeutic gel (ultrasonic gel and hospital hand creme).

Data analysis

EMG data were full-wave rectified and low pass filtered at 50 Hz (6th order Butterworth filter). The onset of EMG activity was identified by computer using an algorithm that selected the point at which the EMG signal deviated by more than 3 standard deviations, for a minimum of 25 ms, above baseline (~200 ms prior to movement) [14]. All onsets were confirmed by visual inspection.

The relative difference in time of onset of VMO and VL EMG was quantified by subtraction of the onset of VMO from that of VL and averaged over the 10 trials. Reaction time was determined as the latency between the stimulus and the onset of the prime-mover for the movement (i.e. TA for the rock task, Sol for the rise task).

Statistical analysis

As data were normally distributed, unpaired t-tests were used to assess baseline comparability in the rock and rise tasks for (i) EMG onset timing differences between VMO and VL, and (ii) reaction time between experimental groups.

Difference scores were calculated by subtracting the baseline scores from the follow-up scores for; (i) the EMG onset timing difference between VMO and VL in the rock and rise tasks, (ii) reaction time in the rock and rise tasks, (iii) the VAS scores, and (iv) the AKP. Separate repeated measures ANOVA's were used to compare the EMG onset timing difference and reaction times between tasks and treatment groups. Independent t-tests were used to compare the difference scores for the VAS and AKP scores between groups. At baseline and at follow up, independent one group t-tests were used to determine if the EMG onset timing differences in the rock and rise tasks differed significantly from zero (i.e., did VMO and VL onset occur simultaneously). The alpha level was set at 0.05.

Results

EMG onset timing difference

Seven subjects (placebo 2, physical therapy 5) did not return for reassessment. There were however no differences between the scores of these seven subjects and the group means. There were no differences at baseline in the rock and rise tasks between groups for EMG onset timing differences between VMO and VL, nor reaction time (all $p > 0.05$). At baseline the onset of VL occurred before VMO for both the placebo and physical therapy intervention groups in both the rock ($p < 0.0001$) and rise ($p < 0.0001$) tasks.

Following the six week treatment program the physical therapy group demonstrated greater change in their EMG onset timing difference of VMO compared to VL than the placebo group ($p < 0.005$) (Figs. 1 and 2). There was no difference in the size of the change between the rock and rise tasks for either group ($p = 0.19$) and no interaction effect between the tasks or groups ($p = 0.22$). In the physical therapy group this change resulted in simultaneous onset of VMO and VL during the rock and rise tasks ($p = 0.75$ and $p = 0.79$, respectively) (Fig. 3a and b). However in the placebo group the EMG onset of VL still occurred before that of VMO in

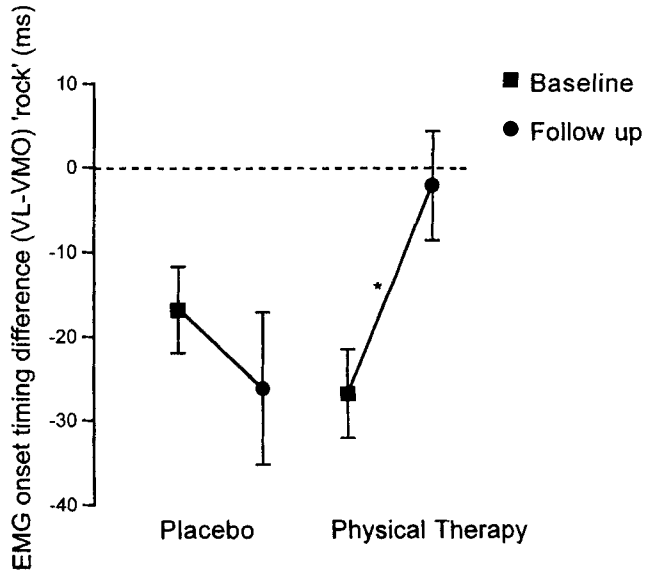

Fig. 1. Means and standard errors for EMG onset timing difference at baseline and following the six week treatment program for the physical therapy and placebo groups in the rock task. While there was no difference in EMG onsets between groups before treatment, only the physical therapy group had a significant improvement in VMO timing relative to VL after treatment (*$p < 0.005$).

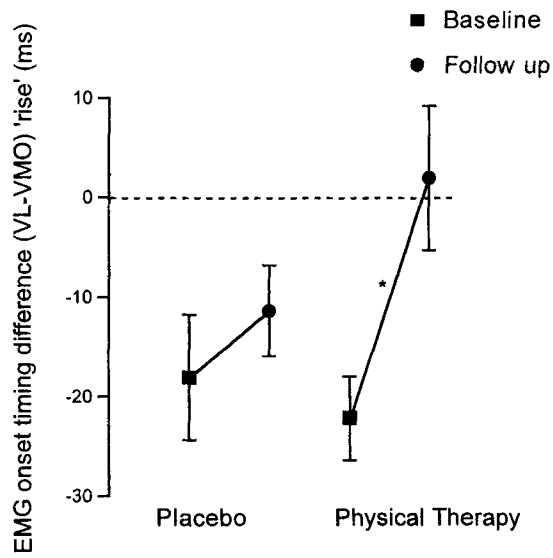

Fig. 2. Means and standard errors for EMG onset timing difference at baseline and following the six week treatment program for the physical therapy and placebo groups in the rise task. While there was no difference in EMG onsets between groups before treatment, only the physical therapy group had a significant improvement in VMO timing relative to VL after treatment (*$p < 0.005$).

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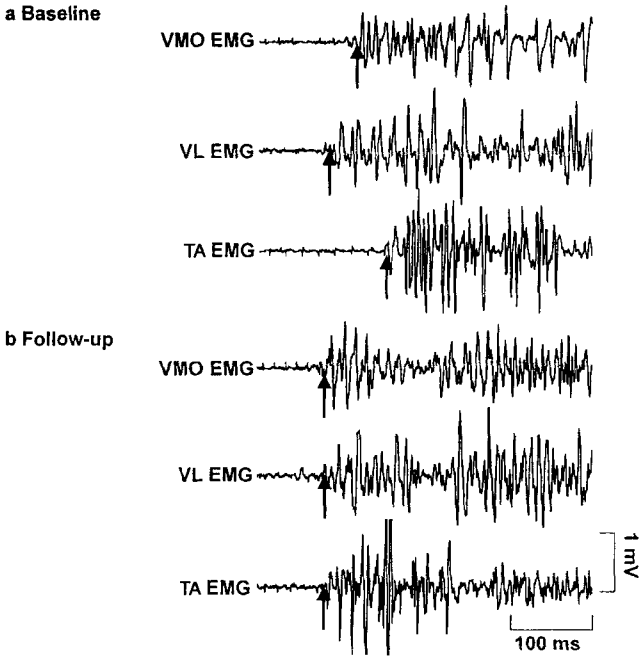

Fig. 3. (a and b) Raw EMG data from a representative subject in the physical therapy group in the rock task before and after intervention. Onsets of EMG activity are indicated with arrows. Note that at baseline the EMG onset of VL occurs prior to that of VMO and that following placebo intervention there is no change in this order of onset.

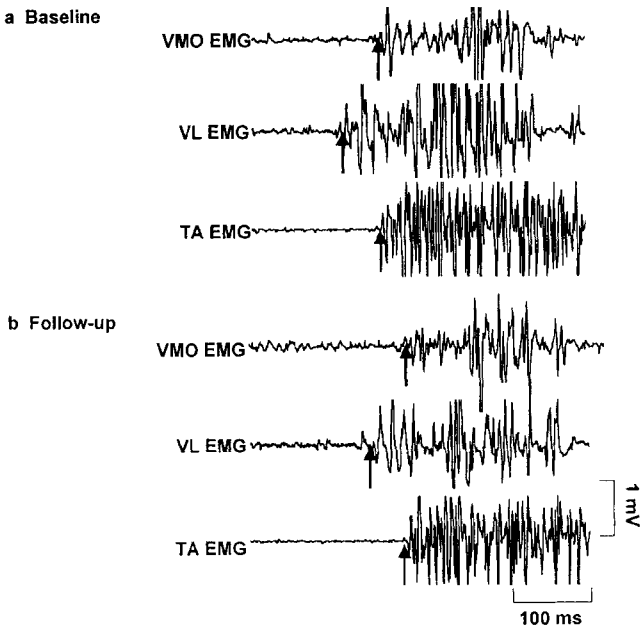

Fig. 4. (a and b) Raw EMG data from a representative subject in the placebo group in the rock task before and after intervention. Onsets of EMG activity are indicated with arrows. Note that at baseline the EMG onset of VL occurs prior to that of VMO whereas following physical therapy intervention the onsets of VMO and VL are virtually simultaneous.

both the rock and rise tasks at the follow-up assessment ($p < 0.05$ and $p < 0.05$, respectively) (Fig. 4a and b).

Pain and disability measures

Improvements in pain and function were greater in the physical therapy than the placebo group. The physical therapy group had significantly greater change in their worst pain in the last week ($p < 0.004$) and the showed greater functional improvement (measured with the AKP) ($p < 0.03$). The data for the larger subject group are reported in full in Crossley et al. [7].

Discussion

This study demonstrates that a physical therapy intervention can change feedforward motor control in a functional task. Specifically, the physical therapy treatment program was found to alter the EMG onset of VMO relative to VL more so than a placebo treatment in people with PFPS. This change in muscle recruitment restored the co-ordination of the medial and lateral components of the vasti to a strategy that was similar to that in a normal population and was associated with a positive clinical outcome. This data provides evidence that activation of the vasti in a feedforward postural strategy can be changed even though this was not specifically addressed in the physical therapy intervention program.

At baseline the EMG onset of VL occurred prior to the onset of VMO in both groups, which concurs with our previous findings [4,6]. Hypothetically, unsynchronised VMO and VL activity may result in patellar tracking dysfunction. Theoretical studies indicate that as little as a 5 ms delay in VMO activation is associated with a significant increase in lateral patellofemoral joint loading [25]. This unsynchronised activity of VMO and VL is in contrast to the virtually simultaneous contraction of the vasti found in the normal population in tasks similar to that used here [5] and other functional tasks such as stair stepping, walking and ascending/descending ramps [4,28].

Specific retraining alters feedforward vasti recruitment

Following the six week treatment program the recruitment of the vasti had been altered in the physical therapy group to great extent than the placebo group. At follow up the specific findings were; firstly in the physical therapy group the EMG onset of VMO occurred at the same time as VL as participants completed both the rock and rise tasks and secondly in the placebo group the EMG onset of VL preceded VMO in both the rock and rise tasks. Feedforward strategies are altered in both neurological conditions (e.g. parkinsons disease [20,30],

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frontal lobe trauma [27]) and musculoskeletal conditions (e.g. delayed onset of transversus abdominis in low back pain [15,16] and VMO–VL in PFPS [6]). However, there is little evidence that these strategies can be altered nor is there consensus on which exercise methods can promote such a change. While several studies have reported improvement in symptoms due to specific motor training [13,26] only one study has provided evidence that this is associated with changes in recruitment of the trained muscles in a functional task that was included in the training program [2]. The present data show that this improvement is transferred to feedforward strategies during tasks that were not trained. Feedforward adjustments are coordinated by the CNS prior to the initiation of the movement and are dependent on the internal model of body dynamics that is developed over the lifetime from movement experience [9]. On the basis of this model it is thought that the CNS predicts the outcome of a movement and plans an appropriate strategy to overcome the perturbation produced by the movement. The present data suggest that this strategy, while altered in people with PFPS, can be restored by a physical therapy program that includes voluntary activation of the VMO in a range of tasks. While the mechanism for the treatment strategy to produce this change cannot be established from this study, the data confirm that planning is changed. This data deals with the 'normal' brain so it is unknown whether the same change would be effected in CNS damaged by neurological disease.

Altered feedforward strategy is associated with a positive clinical change

The change in the relative timing of VMO compared to VL following six weeks of training in the physical therapy intervention group was associated with a positive clinical change (decrease in pain and improvement in function). Whilst other trials have investigated physiotherapy treatment of PFPS [1,8,11,18,24,31,32] none employed a physical therapy program that included specific exercise strategy for retraining VMO. The efficacy of the physical therapy program may lie in the motor relearning basis of the retraining of VMO. That is the use of biofeedback, visual cues, mental imagery, verbal and kinaesthetic feedback, to gain perception of the skill and the repeated practice of the skill and gradual incorporation of the skill into functional activities. However, the use of specific muscle retraining was not the only component of our treatment program. The other components of the treatment program (patellar taping, stretching lateral retinaculum, hamstrings and anterior hip structures) may have improved proprioceptive feedback to the patellofemoral joint and improved biomechanical function of the lower limb or decreased the pain associated with the condition.

Methodological considerations

There are several methodological issues that warrant consideration. The multifaceted treatment program (chosen as it reflects current clinical practice) used makes it impossible to determine which component (or combination of components) was responsible for the change in EMG onset timing difference. Additionally, the study lacked a controlled follow up period and thus no conclusions can be drawn about the long term effects of physical therapy. Seven participants were lost to follow up (five in the physical therapy group and two in the placebo), but analyses of best case (participants received the largest EMG onset timing change) and worst case (participants received the smallest EMG onset timing change) scenarios did not alter the results.

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